Thomas Diederen

Computational biologist

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Summary

I am looking for challenges at the interface of biology and machine learning. I bring experience in simulation based inference, blending modern generative modeling with scientific simulation. My background in biology has honed my ability to navigate large, complex and messy data-sets.

Experience

Consultant, computational biology — Hybrid, Munich, Bavaria
Differential Bio — Jul. 2025 - Jan. 2026 (contract)

- I am developing core machine learning capabilities. We explore hybrid-mechanistic ordinary differential equation (ODE) models + variational inference as well as neural ODE models of biomass & product/substrate time-courses in bio-production processes (implemented in JAX).
- Automated genome scale metabolic model reconstruction and curation pipeline; this project resulted in the first commercial software license for Differential Bio.

Personal goal pursuit — Mediterranean coast
Traveling & climbing — Feb. 2024 – Dec. 2024

Ph.D. titled: Simulation-based Metabolic Flux Inference — Zürich, Switzerland
Zamboni lab, Institute for Molecular Systems Biology, ETH Zürich — Aug. 2018 – Jan. 2024

- Pioneered methods for modeling Riemannian continuous normalizing flows on convex polytopes (paper link).
- Developed pipelines for Bayesian inference of biochemical reaction rates from mass-spectrometry data.
- Engineered the `sbmfi` package to support simulation-based metabolic flux inference (implemented in PyTorch).

Skills

Modelling	Riemannian normalizing flows (discrete and continuous) + flow matching, diffusion models, variational auto-encoders, Bayesian neural networks, Markov-Chain Monte-Carlo, Approximate Bayesian Computation, convex optimization (interior point methods), design of experiments
Programming	Python (advanced), Matlab, JavaScript, R, C++, Bash, PowerShell
Packages	modeling (PyTorch + Lightning, JAX, Ray Tune, Numpy, Scipy, scikit-learn, CvxOpt), visualisation (HoloViews, Matplotlib, Bokeh), data-wrangling (Pandas, Xarray, NetCDF, HDF5), biology (COBRAPy, ModelSEED, KBase, emzed3, SkyLine, Mass Hunter, OpenMS)
General	Git, GitHub, Docker, Cursor, LSF-cluster, LaTeX, Microsoft Office
Languages	Dutch (fluent), Polish (fluent), English (fluent), German (good, B2)

Education

M.Sc. Biotechnology - Systems Biology — Basel, Switzerland
Department of Biosystems Science and Engineering, ETH Zürich — Sept. 2015 - Aug. 2017

- Thesis title: Insulation in the Stochastic Setting. Control Theory and Systems Biology Laboratory
- Courses in computational systems biology, machine learning, linear algebra, data science and control theory

B.Sc. Life Science & Technology — Delft, The Netherlands
Delft University of Technology — Sept. 2011 - Nov. 2014